

NUVO ENERGY

GENERATOR INTEGRATION KIT-100S

DATASHEET Rev 1.0



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1 Overview

1.1 NE-GIK-100

The NE-GIK-100 is a PV-plant control and monitoring system that can be used in conjunction with backup generators. The kit is preassembled and wired with all the needed parts to enable control on any standard PV plant installation.

The system is compatible with all popular inverter brands and supports MODBUS RTU and TCP for controlling/monitoring inverters. The control system can implement grid-feed prevention or grid-export setpoints while connected to grid.

The generator integration is agnostic of generator manufacturer by using a power meter to measure the generator production and adjusting the PV production to enable the generator to run in an optimal condition. The generator integration enables fuel saving by utilizing the PV power to reduce the generator load.

The NE-GIK-100 kit is easy to install and commission and operates autonomously once commissioned.

1.2 NE-GIK-100S

The NE-GIK-100S is a software variant of the NE-GIK-100 specifically meant for Solis inverter installation. It has additional software features and configuration menus to ease the process of commissioning a Solis solar plant with generator integration.

1.2.1 NE-GIK-100S compatibility

The NE-GIK-100S is compatible with all 4G and 5G Solis grid-tie inverters that have an RS-485 interface available for monitoring and control. Canadian Solar inverters that are based on the Solis 4G and 5G inverters may also be used.

The Solis LAN and Wi-Fi expansion cards are a known cause of communication interruption as they are internally connected to the inverter RS-485 communication bus. It is recommended to remove these expansions before integrating the inverters with the NE-GIK-100S kit.

1.3 Features & Benefits

- Easy to install
- Web based portal accessible from your PC, phone or tablet
- Zero-feed or export capabilities while on grid
- Works with any manufacturer or size generator
- Industry compliant
- Generator fuel saving
- Solar irradiance sensor monitoring
- Check meter monitoring

1.4 Typical applications

- Solar Energy Monitoring
- Grid-Tie inverter performance monitoring
- Generator integration with solar

2 Specification

2.1 Electrical specification

	Characteristic	Value
Electrical	Voltage Input (Load)	3P&N, 100-277 V L-N
	Voltage Input (Genset)	1P&N, 100-277 V L-N
	Frequency	50 / 60 Hz $\pm 10\%$
	Current Input (CT)	1A or 5A secondary
Cables	RS-485 connection	Shielded twisted pair
	Ethernet connection	RJ45 - 10/100Mbps
	Terminal cable sizes	Max 6mm ²
Comms	RS485	Modbus RTU
	Ethernet 1 (Internet feed)	RJ45 10/100Mbps with DHCP
	Ethernet 2 (Inverter Comms)	Modbus TCP

2.2 Mechanical specification

	Characteristic	Value
Enclosure	Index Protection (IP)	IP65
	Description	Wall mount sheet metal enclosure
	Colour	RAL2000 (electric orange)
	Dimensions (mm)	(H) 550 x (W) 400 x (D) 220